B.Tech CSE - Third Year | Lab 14

Assignment 14: Sorted Check, Search and Fibonacci • *Aim: Use recursion for sorted checks, searching and Fibonacci values.*

Experiment No. 14.1: Recursively check sorted array

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Recursively_check_sorted_array {
    static boolean isSorted(int[] a, int i) {
        if (i >= a.length - 1) return true;
        if (a[i] > a[i + 1]) return false;
        return isSorted(a, i + 1);
    }
    public static void main(String[] args) {
        int[] a = {2, 4, 6, 8, 10};
        System.out.println(isSorted(a, 0));
    }
}
```

3. **Output:** `true` (Time: O(n), Space: O(n) stack space)
4. **Viva-Voce:** What does ascending order mean? Why compare adjacent elements? When does the method return false?

Experiment No. 14.2: Find index recursively

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Find_index_recursively {
    static int findIndex(int[] a, int key, int i) {
        if (i == a.length) return -1;
        if (a[i] == key) return i;
        return findIndex(a, key, i + 1);
    }
    public static void main(String[] args) {
        int[] a = {45, 67, 80, 52, 90};
        System.out.println(findIndex(a, 80, 0));
    }
}
```

3. **Output:** `2` (Time: O(n), Space: O(n) stack space)
4. **Viva-Voce:** What does -1 mean? Does recursive search require sorted input? What is returned when found?

Experiment No. 14.3: Calculate Fibonacci recursively

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Calculate_Fibonacci_recursively {
    static int fib(int n) {
        if (n == 0) return 0;
        if (n == 1) return 1;
        return fib(n - 1) + fib(n - 2);
    }
    public static void main(String[] args) {
        System.out.println(fib(7));
    }
}
```

3. **Output:** `13` (Time: $O(2^n)$, Space: O(n) depth)
4. **Viva-Voce:** What are the first two Fibonacci values? What is the recurrence relation? Why is this version inefficient for large n?